NOVA IMS Information Management School

MATHEMATICAL ANALYSIS 2

TEST 3
Fall semester 2024/25
January 13, 2025 - 1 pm

Student Name: Student Number:

Instructions:

- The time limit is 1 hour.
- Clearly justify your answers and present all your calculations.
- Respect the answer space given for each question.
- The draft sheet is stapled at the end of the test. Do not remove it.
- Use only a blue or black ink pen. Answers written in pencil will not be graded.
- It is not allowed the use of calculators.
- It is not allowed the use of any study notes with the exception of the authorized formulae sheet. You cannot write on the formulae sheet.
- The possession, handling or display of cell phones, smart watches, or any other digital devices is not allowed, and it is sufficient reason for the annulment of the test, regardless of its use.

Grade:________________

1. Consider the function g defined by

$$g(x, y) = y\sin(x) + x^2 + y^2$$

[1.5] (a) Prove that (0,0) is a critical point of g.

[2.5] (b) Classify the critical point (0,0) as a maximum, a minimum or a saddle point.

ANSWER

[3.0] 2. Consider the function f defined in $\mathbb{R}^2$ by:

$$f(x, y) = e^x \, y$$

Use Lagrange Multipliers Method to find the extreme values of f when subject to the constraint $x + y = 0$.

ANSWER

3. Consider the following subset of $\mathbb{R}^2$:

$$A = \left\{(x,y) : y \leq \sqrt{x} \wedge y \geq x - 2 \wedge y \geq 0\right\}$$

[1.5] (a) Draw the region defined by the set A.

[2.0] (b) Using double integral(s) compute the area of A.

[3.0] (c) With a graphical approach and a detailed explanation, find the minimum and the maximum values of $h(x,y) = (x-4)^2 + (y-1)^2$ on the set A.

ANSWER

4. Consider the following double integral:

$$\int_0^1 \int_{-1}^{x} \frac{2xy}{(y^2+1)^2}\, dy\, dx$$

[1.5] (a) Draw the region of integration.

[2.0] (b) Interchange the order of integration.

[3.0] (c) Compute <u>the given</u> double integral.

<u>ANSWER</u>

End

Draft Sheet